

DATA PARALLELISM: HOW TO TRAIN DEEP LEARNING MODELS ON MULTIPLE GPUS

LAB 3, PART 1: SCALING THE BATCH SIZE



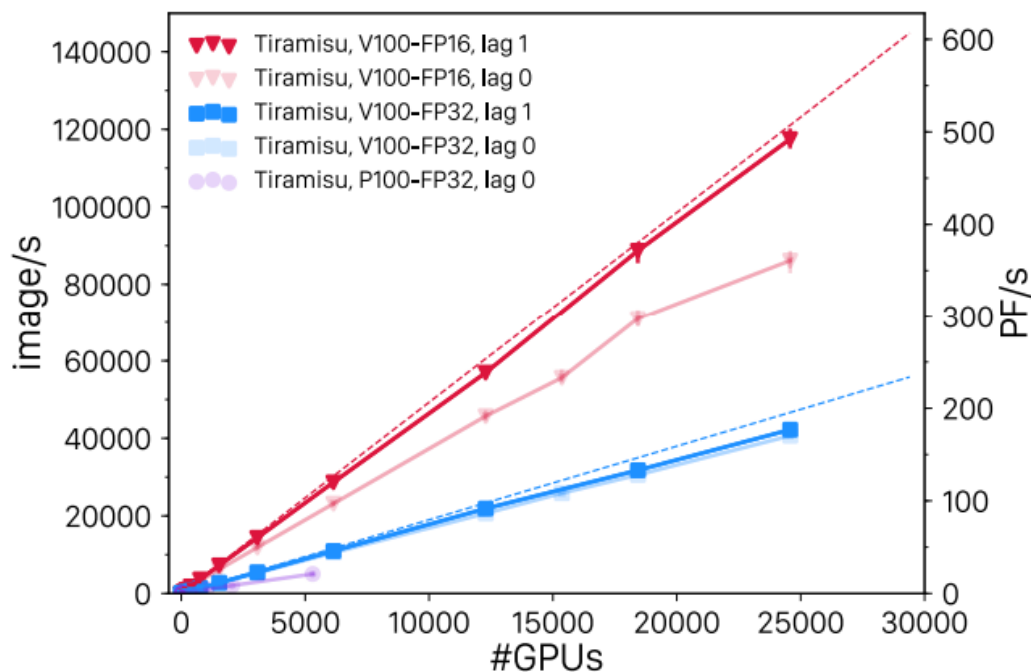
DEEP
LEARNING
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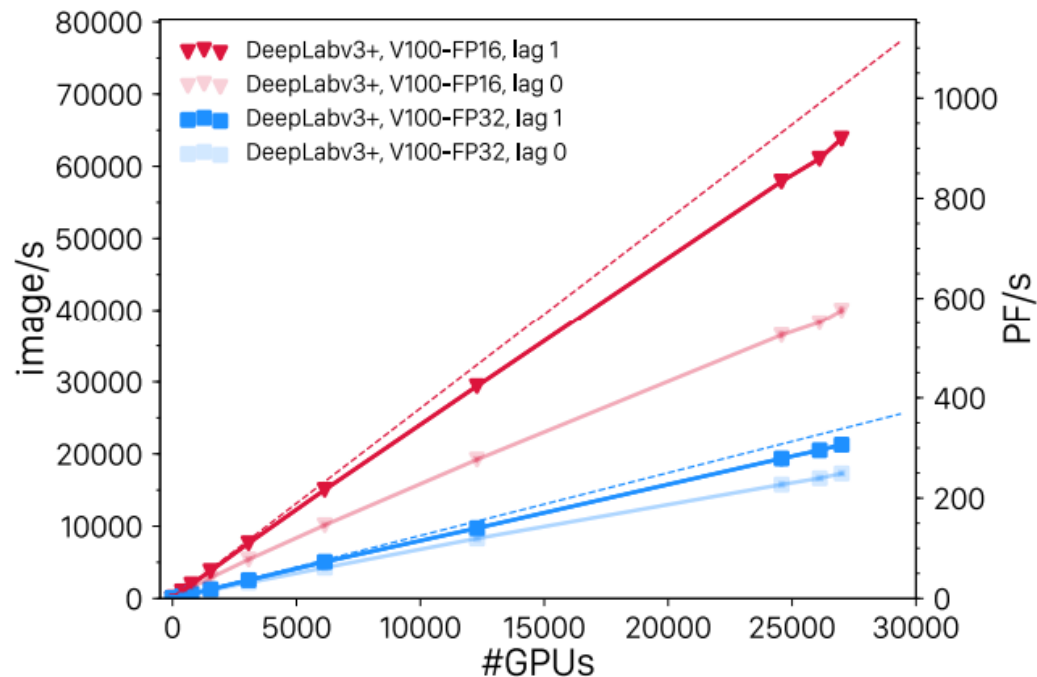
**CAN WE INCREASE THE BATCH SIZE
INDEFINITELY?**

IN TERMS OF IMAGES / SECOND?

Yes



(a) Tiramisu

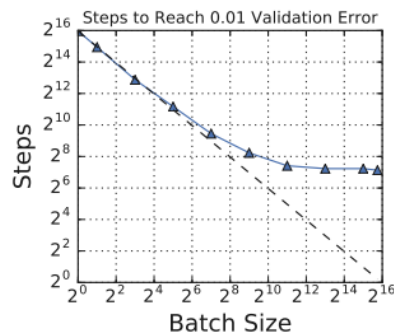


(b) DeepLabv3+

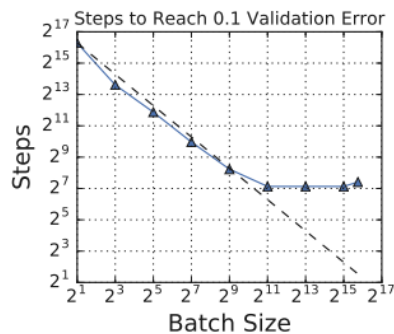
Kurth, T., Treichler, S., Romero, J., Mudigonda, M., Luehr, N., Phillips, E., ... & Houston, M. (2018, November). Exascale deep learning for climate analytics. In Proceedings of the International Conference for High Performance Computing, Networking, Storage, and Analysis (p. 51). IEEE Press. [arXiv:1810.01993](https://arxiv.org/abs/1810.01993)

IN TERMS OF STEPS TO CONVERGENCE?

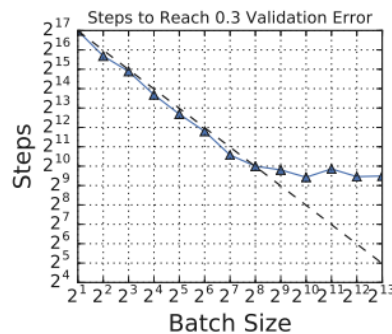
There are limits



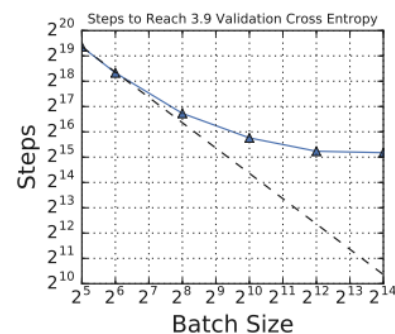
(a) Simple CNN on MNIST



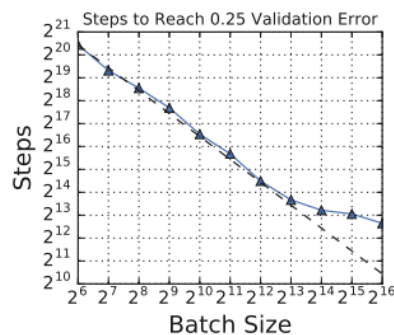
(b) Simple CNN on Fashion MNIST



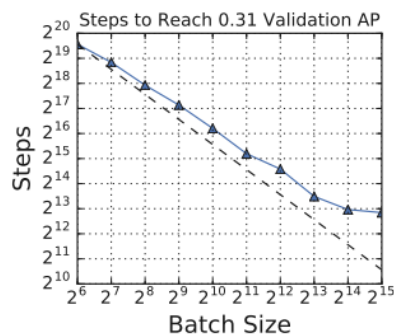
(c) ResNet-8 on CIFAR-10



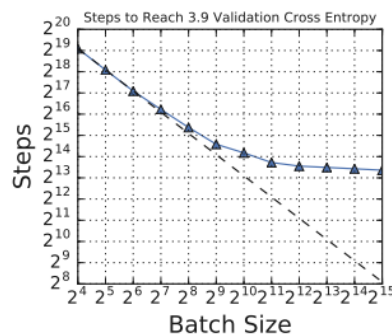
(g) Transformer on Common Crawl



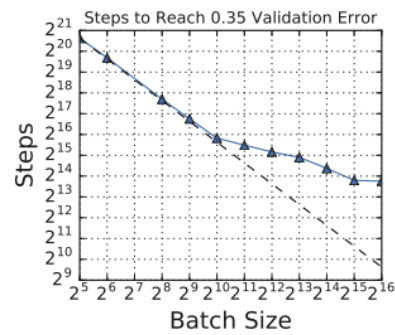
(d) ResNet-50 on ImageNet



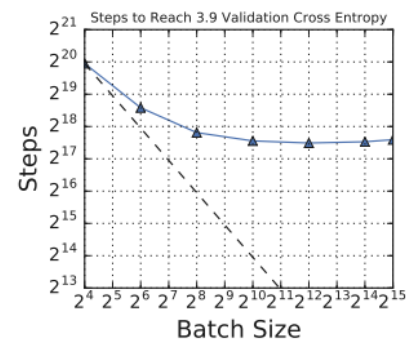
(e) ResNet-50 on Open Images



(f) Transformer on LM1B



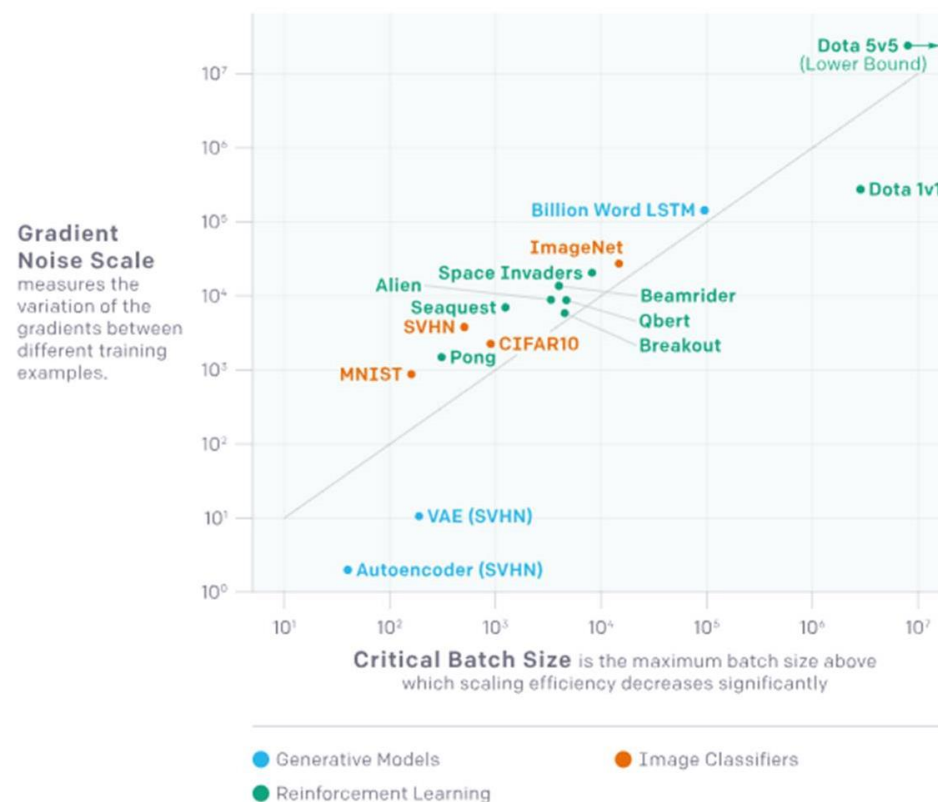
(h) VGG-11 on ImageNet



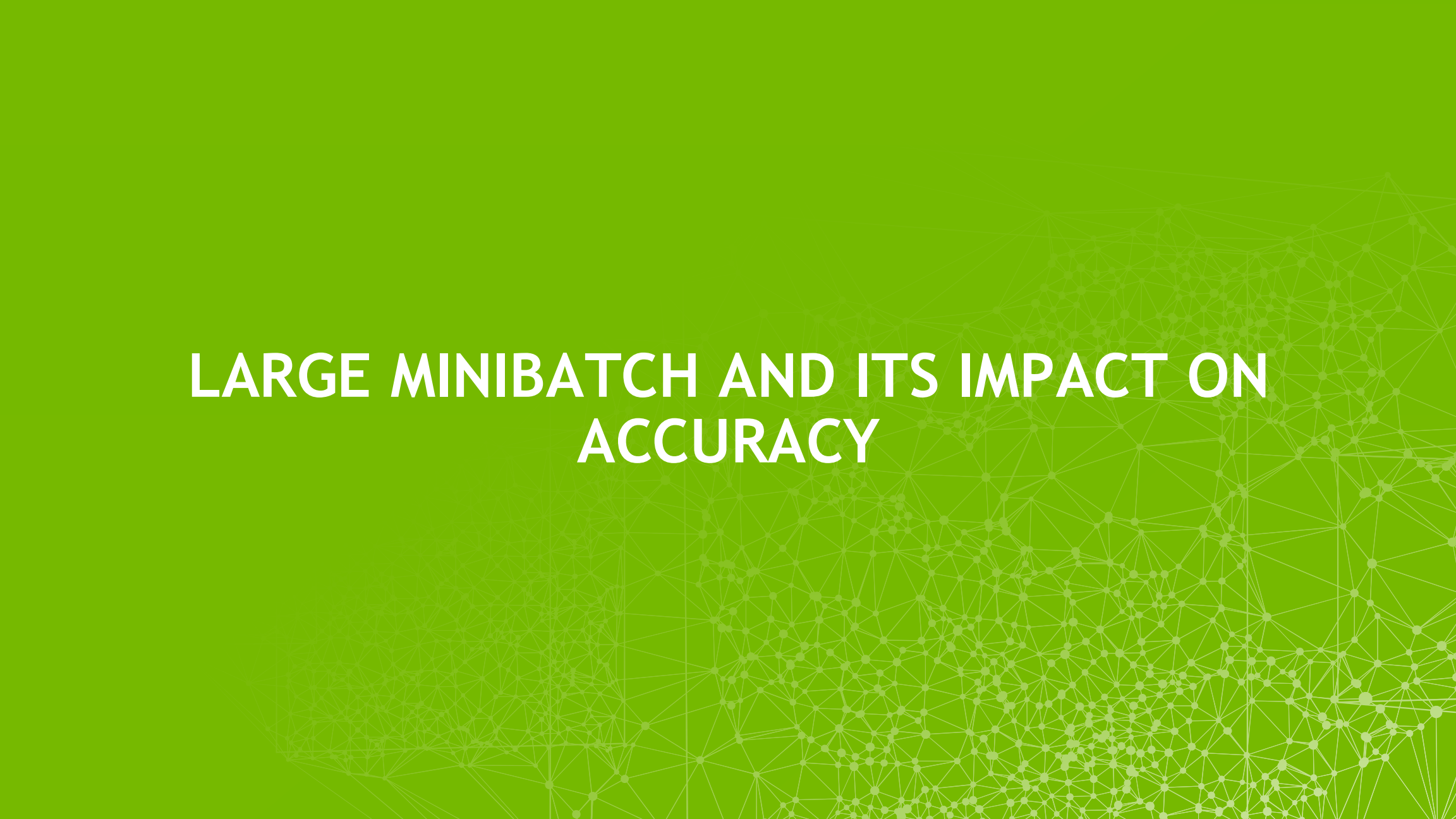
(i) LSTM on LM1B

IN TERMS OF STEPS TO CONVERGENCE?

There are limits

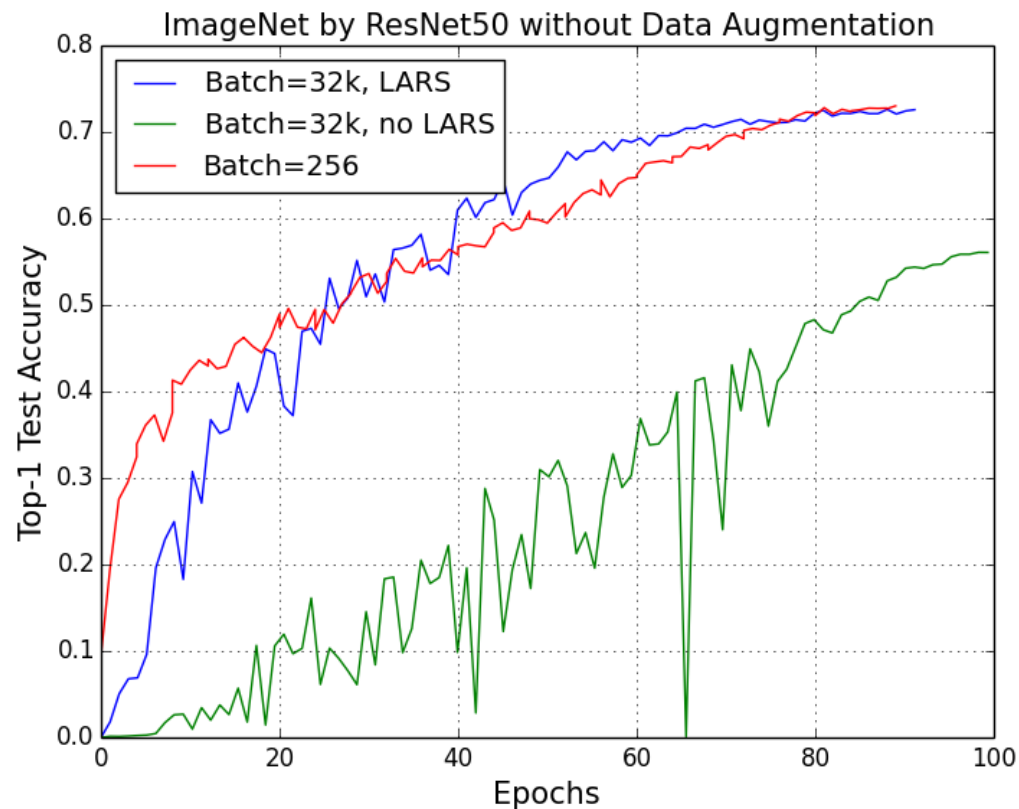
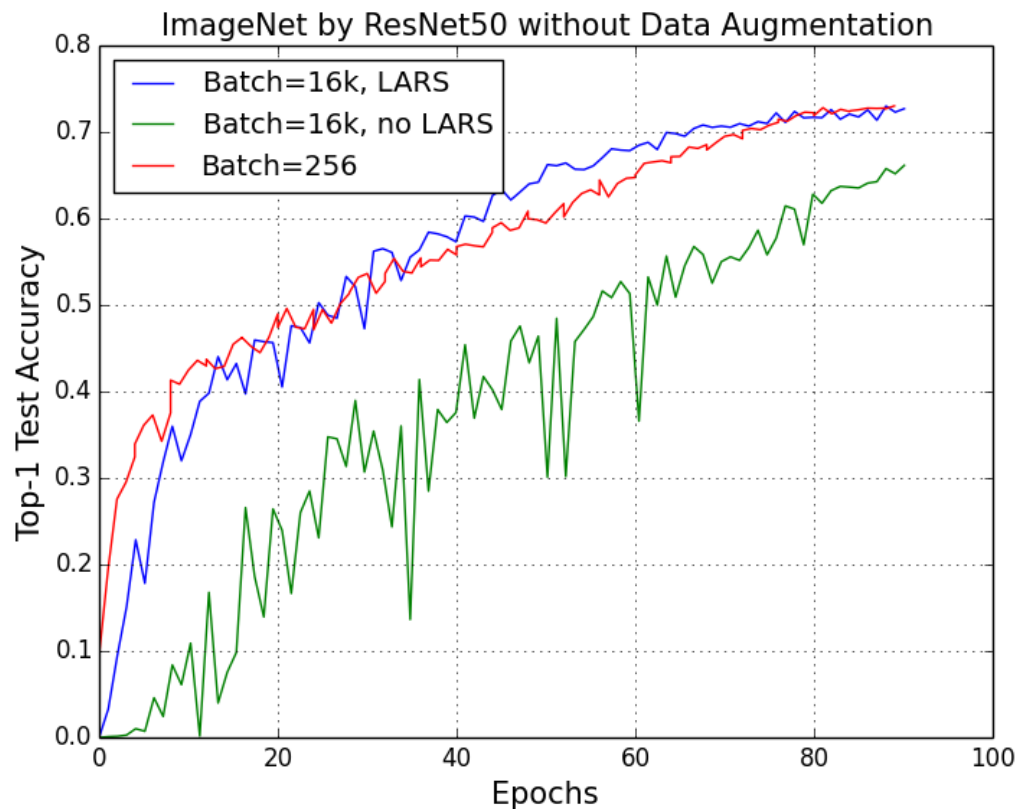


LARGE MINIBATCH AND ITS IMPACT ON ACCURACY

The background of the slide features a smooth color gradient transitioning from a deep green on the left to a bright yellow on the right. Overlaid on this gradient is a complex, abstract network of small white dots connected by thin white lines, creating a mesh-like pattern that resembles a neural network or a data structure. The density of the network increases towards the right side of the slide.

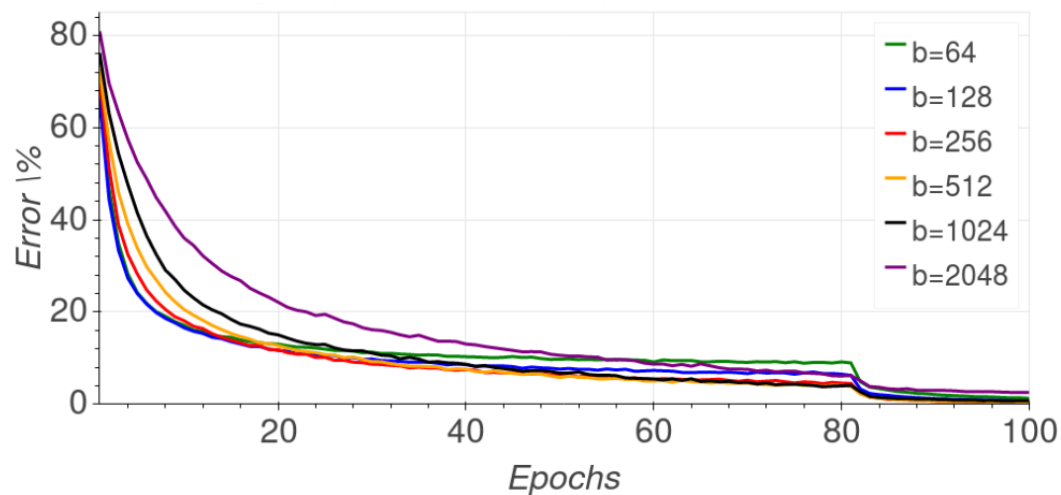
IMPACT ON ACCURACY

Naïve approaches lead to degraded accuracy

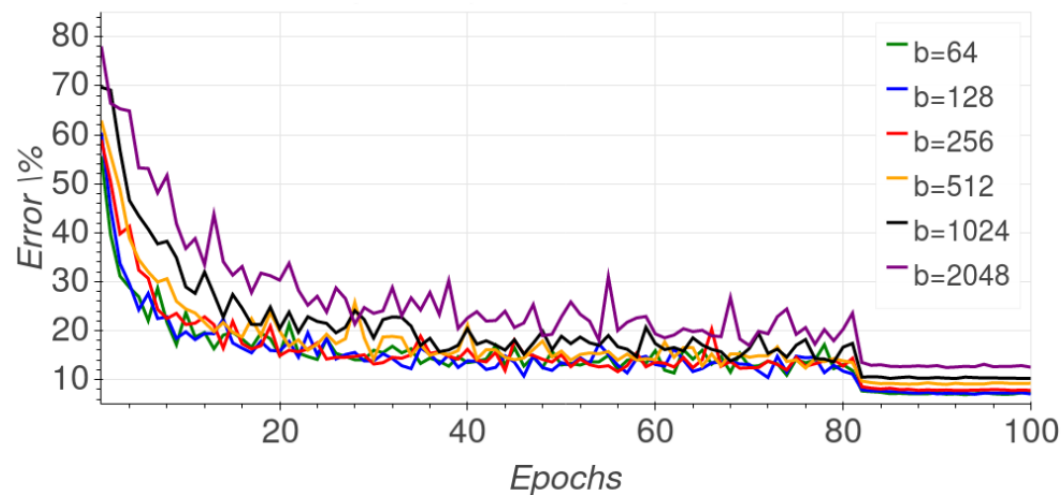


IMPACT ON ACCURACY

Naïve approaches lead to degraded accuracy



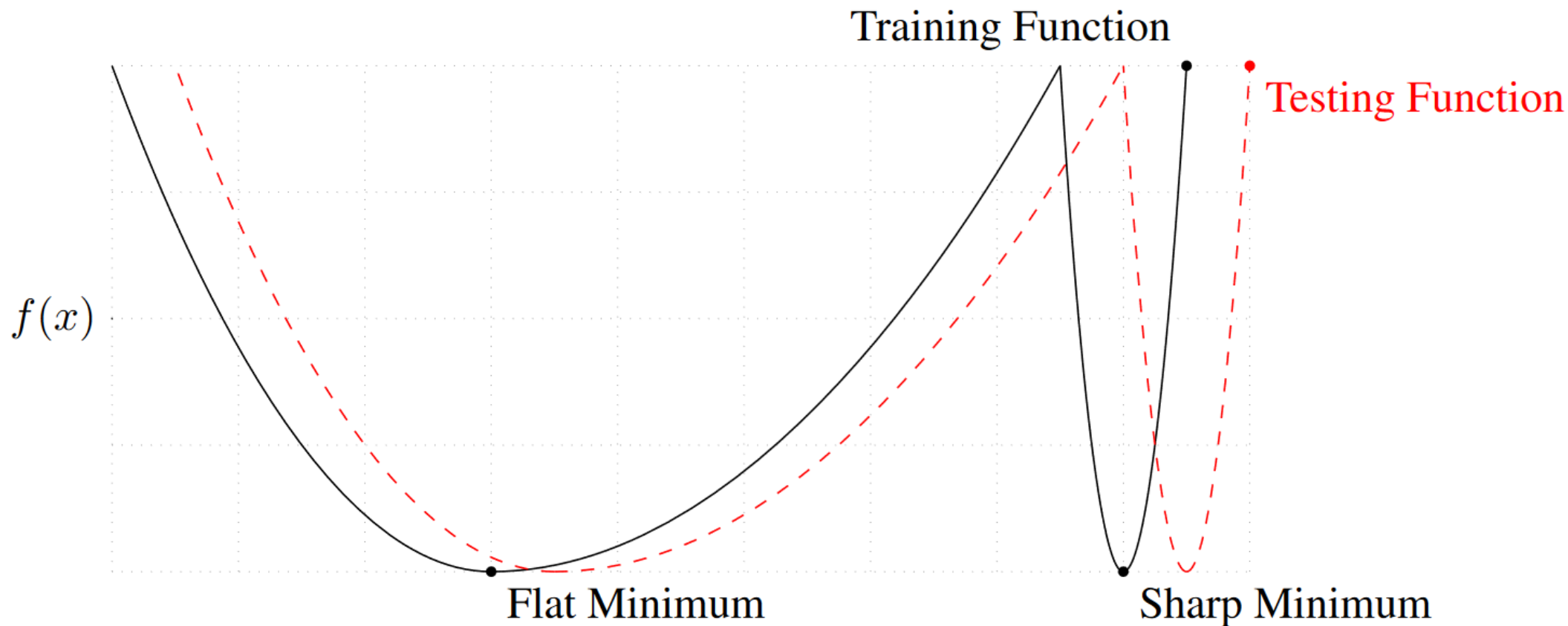
(a) Training error



(b) Validation error

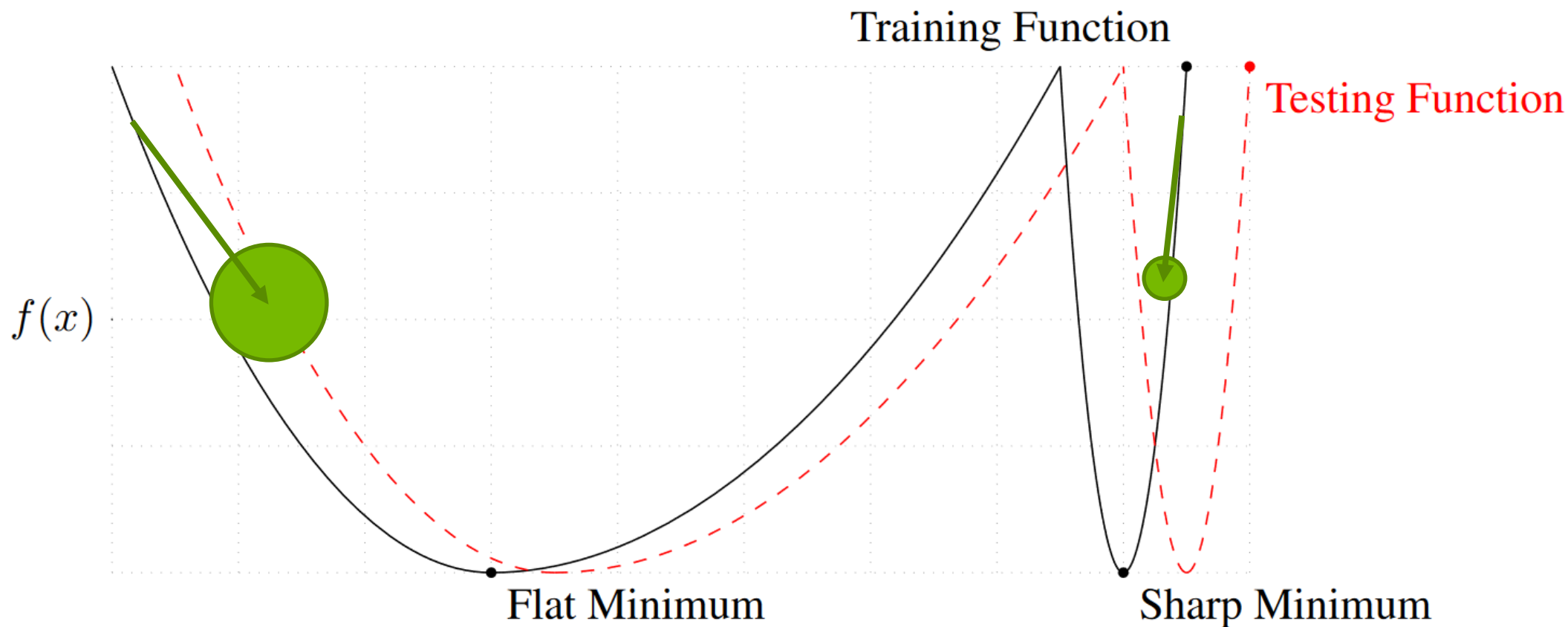
IMPACT ON ACCURACY

Why? Generalization and flatness of minima?



IMPACT ON ACCURACY

Why does it happen? Noise in the gradient update.



IMPACT ON ACCURACY

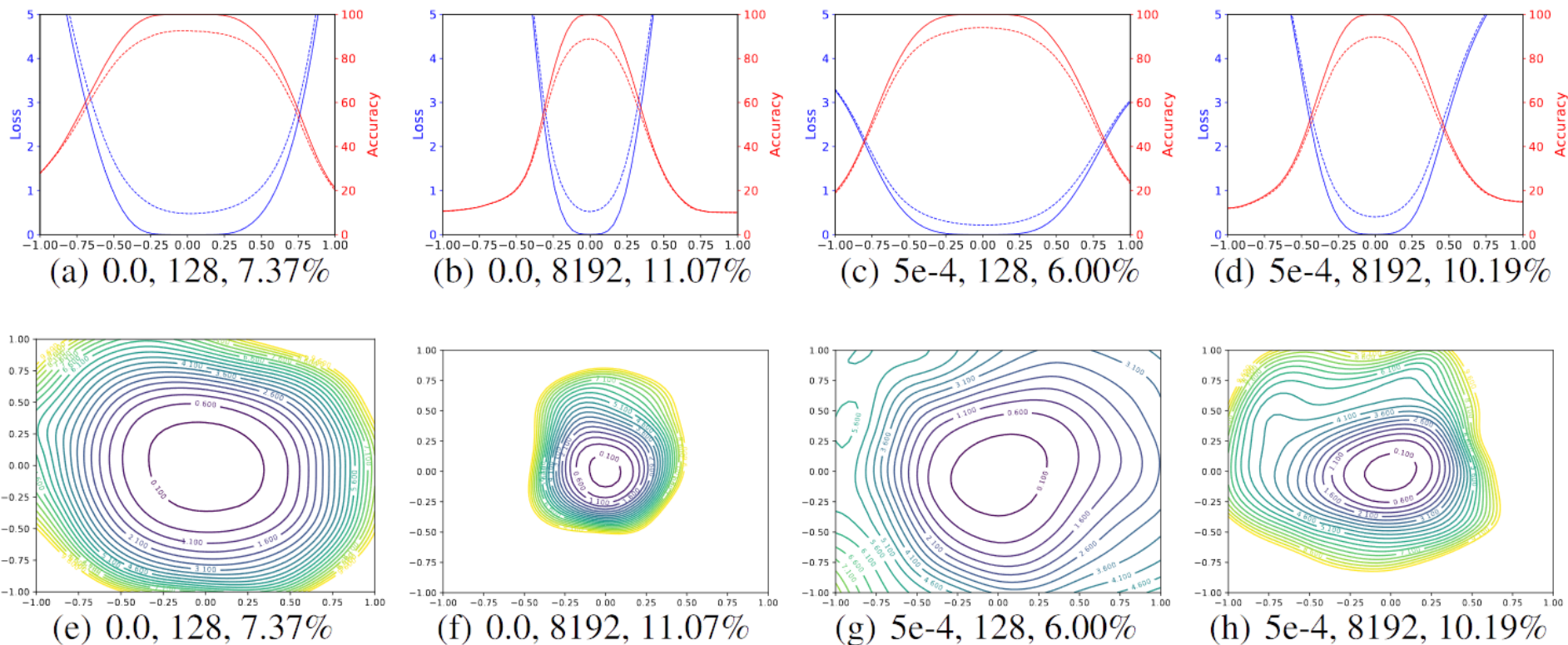
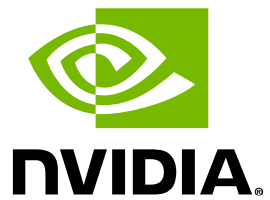


Figure 3: The 1D and 2D visualization of solutions obtained using SGD with different weight decay and batch size. The title of each subfigure contains the weight decay, batch size, and test error.



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